

## LUMAFLOW · CALCULATION

# LumaFlow sizing calculations

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## DOCUMENT CONTROL

|             |                                       |         |                |
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| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/lumaflow |         |                |
| Source      | docs/04-calcs/01-sizing.md @ 9904b27  |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                                    |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); rerun for the 50 mm bore, 0.95 liner, 1.2 L/min, wall dose sensor, 15 Hz per L/min flow sensor, 10 mm window, M5 rods, pressure limiter and thermal cut-back |
| 0.1     | 2026-09-25 | Amish Chadha | First issue for TRL 3 (flow, Monte Carlo optics and dose, dose monitor, switching, power, thermal, pressure drop, window and structure, size, cost, options)                                              |

## LumaFlow sizing calculations

This note checks the LumaFlow design as revised by LMF-DDR-002: a 50 mm bore lined with high-reflectance PTFE, a design flow of 1.2 L/min (0.32 gpm), a UV-C photodiode in the tube wall at mid height, a flow sensor of 15 Hz per L/min, an upstream pressure limiter and a thermal cut-back. On paper, LumaFlow now meets thirteen of its seventeen requirements, has one at risk (R4) and misses two (R3 and R16); one cannot be verified at TRL 3. The reduction equivalent dose (RED) in clear water (90 %/cm UV transmittance, UVT) is 51.9 (laminar) to 76.3 (plug) mJ/cm<sup>2</sup>, so **R2 is now met** with the 20 % margin this note requires; in v0.1, with a 25 mm bore at 2.0 L/min, it was 14.7 to 19.0 mJ/cm<sup>2</sup>. The dose at the 70 %/cm test condition is 17.4 to 21.8 mJ/cm<sup>2</sup> (**R3 not met**, kept visible by decision D1), and the parts cost \$338.00 against the \$225 budget (**R16 not met**). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B5], is the line of that script's output that carries it.

**Safety:** These calculations concern a device meant to disinfect drinking water with UV-C light under mains pressure. They are first-principles estimates for a paper proof of concept, not a substitute for biodosimetry, datasheets or a review by a qualified engineer. A calculated dose above target is not a demonstrated dose. LumaFlow is a research and educational prototype, not a certified water treatment device. See LMF-PRC-001, Safety.

### ■ Scope and method

The note checks every requirement in LMF-REQ-001 v0.4 against the design in LMF-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `levels()` and part volumes, so the dimensions used here are the ones in the STEP files and in drawing LMF-DWG-001 Rev P2. It reads prices from `bom/bom.csv` and the budget from `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py` (about 20 s; it runs the ray trace eight times).

Status rule for the dose: the laminar-flow RED is the design value, because the laminar profile is the more pessimistic of the two limits and the real profile lies between them [A5]. **Met** needs the laminar RED to clear the target by 20 %.

### ■ Assumptions

*Table 1. Main assumptions.*

| Area          | Assumption                                                                                                                                                                                                                                                      | Basis                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Flow          | 1.2 L/min (0.32 gpm), set by the restrictor; water at 20 °C for hydraulics, 25 °C for the thermal case                                                                                                                                                          | LMF-REQ-001 v0.4, R1                                                                  |
| LEDs          | Six 275 nm LEDs on a 32 mm circle, 60 mW UV-C each at 350 mA and 6.2 V (0.36 W, 13.02 W electrical); Lambertian emission                                                                                                                                        | Typical 3535-class UV-C LED ratings; bin to be confirmed                              |
| Window        | UV-grade fused silica, 57 x 10 mm on a 51 mm seat, $n = 1.496$ at 275 nm, bulk transmittance 0.995; air gap 0.8 mm; unpolarized Fresnel losses at both faces; light outside the seat aperture or the 50 mm bore is lost                                         | Standard optics                                                                       |
| Water         | $n = 1.372$ ; base-e absorption $\alpha = -\ln(\text{UVT})$ per cm (0.105/cm at 90 %/cm, 0.357/cm at 70 %/cm); no scattering                                                                                                                                    | UVT definition                                                                        |
| Walls         | Diffuse (Lambertian) reflection. Wetted high-reflectance PTFE liner 0.95 (sensitivity 0.80 and 0.90); wetted 316 stainless in the lower cap bore 0.30; window and LED side seen from the water 0.10                                                             | Engineering judgment for a sintered or expanded high-reflectance PTFE; to be measured |
| Ray trace     | 600,000 rays per case, 20 equal-area annuli by 60 axial bins; fluence rate = absorbed power / ( $\alpha \times$ bin volume); the wall sensor window is an 8 x 8 mm absorbing patch at mid height                                                                | Collision estimator                                                                   |
| Flow profile  | Two limits: plug flow and fully developed laminar (parabolic) flow                                                                                                                                                                                              | Entrance length [A5]                                                                  |
| Dose response | First-order inactivation of a challenge organism with 20 mJ/cm <sup>2</sup> per log (MS2-like), with the same response at 275 nm as at 254 nm                                                                                                                   | Assumption; the wavelength response must be checked                                   |
| Photodiode    | SiC, 0.13 A/W at 275 nm, 0.06 mm <sup>2</sup> active area, behind an 8 mm quartz window of 0.92 transmittance in the tube wall                                                                                                                                  | Typical TO-46 SiC photodiode                                                          |
| Flow sensor   | Hall-effect turbine, 15 Hz per L/min; Kv 0.35 m <sup>3</sup> /h                                                                                                                                                                                                 | Pulse rate set by LMF-DDR-002; Kv to confirm                                          |
| Valve         | 24 V DC normally closed, 4.8 W (0.2 A) held open, Kv 0.25 m <sup>3</sup> /h                                                                                                                                                                                     | Typical small direct-acting solenoid                                                  |
| Electronics   | Driver 90 %; controller 0.6 W active; adapter 88 % at load, 70 % at light load, 0.1 W no-load                                                                                                                                                                   | US DOE Level VI no-load class for small adapters                                      |
| Use           | 15 L/day in 20 draws, 5 s run-on per draw                                                                                                                                                                                                                       | LMF-REQ-001                                                                           |
| Thermal       | Sink natural convection 5 W/(m <sup>2</sup> ·K) (fins hanging down) plus radiation, $\epsilon = 0.85$ ; 0.5 mm pads of 3 W/(m·K); 316 at 16 W/(m·K); water film 300 W/(m <sup>2</sup> ·K) in the lower cap (range 150 to 600); 10 K/W junction to board per LED | Engineering judgment; film coefficient is the main uncertainty                        |
| Structure     | Fused quartz design tensile stress 6.8 MPa (1,000 psi) ( <a href="#">Momentive, mechanical properties</a> ); Poisson ratio 0.17; simply supported edge; 8 bar working pressure; pressure limiter set at 4 bar                                                   | Roark circular plate, uniform load                                                    |

## ■ A. Flow and hydraulics (R1)

The irradiated channel is 50 mm across and 242 mm long, from the window to the PTFE top disc; the flow path from inlet to outlet is 226 mm [A1]. The channel holds 475 mL, 444 mL of it between the ports [A2]. At 1.2 L/min the mean velocity is 1.02 cm/s and the mean residence time 22.2 s [A3]. The Reynolds number is 507 at 20 °C and 334

at 5 °C, so the flow is laminar [A4]. The laminar entrance length, about 1.3 m, is far longer than the flow path [A5], so the velocity profile lies between plug flow and a parabola; both limits are carried through section B. R1 is met by the 1.2 L/min restrictor, and the flow sensor range of 0.3 to 6 L/min covers it.

## ■ B. Optics and UV-C dose (R2, R3)

The six LEDs give 0.36 W of UV-C from 13.02 W electrical, a UV-C efficiency of 2.8 % [B1]. Of that, 87.0 % enters the water [B2]. In the 50 mm bore light crosses twice as much water between wall reflections as in the old 25 mm bore, and the liner returns 95 % of what reaches it. At 90 %/cm the water absorbs 54.0 % of the LED output and the walls 27.0 % (v0.1: 26.1 % and 49.8 %); at 70 %/cm the water absorbs 68.1 % [B3, B4].

*Table 2. Dose at 1.2 L/min (mJ/cm<sup>2</sup>). Average is the flow-weighted average dose; RED is for the 20 mJ/cm<sup>2</sup> per log organism.*

| UVT (%/cm) | Average dose | RED, plug flow | RED, laminar flow | Reactor efficiency, laminar |
|------------|--------------|----------------|-------------------|-----------------------------|
| 95         | 134.3        | 133.7          | 84.5              | 0.63                        |
| 90         | 76.8         | 76.3           | 51.9              | 0.68                        |
| 85         | 51.7         | 51.3           | 36.8              | 0.71                        |
| 80         | 37.5         | 37.2           | 27.7              | 0.74                        |
| 75         | 28.4         | 28.1           | 21.7              | 0.77                        |
| 70         | 22.0         | 21.8           | 17.4              | 0.79                        |

At the design water quality of 90 %/cm the RED is 51.9 (laminar) to 76.3 (plug) mJ/cm<sup>2</sup> [B5]. **R2 (40 mJ/cm<sup>2</sup>) is met**, with the laminar value 30 % above the target. At 70 %/cm it is 17.4 to 21.8 mJ/cm<sup>2</sup> [B6], so **R3 is not met**, as decision D1 accepted; it is now above the Class B level of 16 mJ/cm<sup>2</sup>.

The reactor efficiency (RED over average dose) is 0.68, lower than the 0.77 of v0.1: the slow laminar core on a wide bore sees less light than the water beside the wall. Streamline doses at 90 %/cm range from 42 mJ/cm<sup>2</sup> on the axis to 80 at mid radius and 1,453 beside the wall [B7]. The liner reflectance is the main optical uncertainty: 46.3, 49.9 and 51.9 mJ/cm<sup>2</sup> at 0.80 (plain PTFE), 0.90 and 0.95 [B8], so even plain PTFE meets 40 mJ/cm<sup>2</sup> but not the 20 % margin. LEDs 20 % below rating (bin, aging or temperature) give 41.5 mJ/cm<sup>2</sup> [B9].

## ■ C. Dose monitor (R4)

The photodiode now sits behind an 8 mm quartz window in the tube wall at mid height (Z = 166 mm), facing the enclosure. It sees 0.23 mW/cm<sup>2</sup> and gives about 16.3 nA at 90 %/cm, 2.6 nA at 80 %/cm and 0.38 nA at 70 %/cm [C1], a signal across the whole range; the top-disc position of v0.1 gave 2.4 nA at 90 %/cm and nothing resolvable at 70 %/cm. Between 90 and 85 %/cm the wall signal falls 2.9 times as fast, in log terms, as the RED [C2] (4.4 times for the old position).

*Table 3. Wall sensor signal against UVT.*

| UVT (%/cm) | Irradiance (mW/cm <sup>2</sup> ) | Photocurrent (nA) | Signal relative to 90 % | RED relative to 90 % |
|------------|----------------------------------|-------------------|-------------------------|----------------------|
| 95         | 0.738                            | 53.0              | 3.25                    | 1.63                 |
| 90         | 0.227                            | 16.3              | 1.00                    | 1.00                 |
| 85         | 0.084                            | 6.1               | 0.37                    | 0.71                 |
| 80         | 0.036                            | 2.6               | 0.16                    | 0.54                 |
| 75         | 0.012                            | 0.9               | 0.055                   | 0.42                 |
| 70         | 0.005                            | 0.38              | 0.024                   | 0.34                 |

The controller keeps the proportional mapping  $RED_{est} = RED_{ref} \times (S/S_{ref}) \times (Q_{ref}/Q)$ , which blames every loss of signal on the LEDs and so always under-reads. With  $RED_{ref} = 51.9 \text{ mJ/cm}^2$  the alarm threshold is  $S/S_{ref} = 0.77$  [C3]. With LEDs at rated output the alarm trips below about 88.7 %/cm UVT, and LEDs aged to 77 % of output trip it in 90 %/cm water, where the true RED is  $40.0 \text{ mJ/cm}^2$  [C4]. A mapping that blamed the water instead would be unsafe: with LEDs at 70 % output in 90 %/cm water it would claim  $46.4 \text{ mJ/cm}^2$  when the true RED is  $36.3$  [C5]. R4 stays **at risk** because the sub-nanoampere signal in poor water, the mapping and the 1 s fail-safe still need firmware and a test, which are TRL 4 work.

## ■ D. Flow switching and run-on (R5)

At the 0.3 L/min switch-on threshold the 15 Hz per L/min sensor gives 4.5 Hz, one pulse every 0.22 s [D1]. With the LED rise time the worst-case switch-on is 0.24 s against 0.5 s [D2] (v0.1: 0.46 s with a 7.5 Hz per L/min sensor): **R5 is met**.

During the 5 s run-on the water held in the channel receives a volume-average  $19 \text{ mJ/cm}^2$  at 90 %/cm [D4]. Taking the dose each parcel collected while flowing plus the run-on, every parcel in the flow path has at least  $62 \text{ mJ/cm}^2$  when flow stops (plug flow) [D3].

## ■ E. Power and energy (R9)

The LEDs draw 13.02 W; with the driver, the 4.8 W valve and the controller the 24 V bus carries 19.87 W, and the mains draw is 22.6 W while water flows [E1]. Standby is 0.21 W, dominated by the adapter's no-load draw and the powered flow sensor [E2]. **R9 is met** (25 W and 0.5 W).

At 15 L/day and 1.2 L/min the LEDs run 14.2 min/day (86 h/year); the unit uses 10.4 Wh/day or 3.8 kWh/year [E3]. At that rate 10,000 h of LED on-time would take 116 years [E4], so LED life is set by aging in storage and by switching, not by on-time. For comparison, tap-scale mercury units drawing 13 to 22 W all day (VIQUA VT1, VT4 and S2Q-PA spec sheet) use 114 to 193 kWh/year [E5].

## ■ F. Thermal (R10)

The LEDs turn 12.66 W into heat. The larger 90 x 90 mm finned sink rejects it to cabinet air through 2.46 K/W; the sink, spreader ring, lower cap and board store 802 J/K [F1]. The water path through the spreader ring and the 316 stainless lower cap is 1.55 K/W and carries 9.0 W of the 12.7 W into the water, which warms by 0.11 K [F2].

With continuous flow the LED board settles at 44.8, 40.3 or 37.1 °C for water-film coefficients of 150, 300 or 600 W/(m<sup>2</sup>·K) [F3], below 50 °C in every case; the LED junctions sit near 61 °C [F4]. The board temperature sensor and the cut-back of LED current above 50 °C (decision in LMF-DDR-002) are a backstop for the uncertain film coefficient: **R10 is met**. An acetal lower cap, with no water path, would settle at 63 °C and pass 50 °C after 31.4 min of flow [F5], which supports the 316 lower cap (LMF-DDR-002 confirms it). If the LEDs stuck on with no flow, the board would head for 53 °C, where the cut-back would act; the 5 s run-on adds only 0.08 K [F6].

## ■ G. Pressure drop, window and structure (R7, R8, R17)

Table 4. Pressure drop at 1.2 L/min, excluding the flow restrictor [G1, G2].

| Element                  | Drop (bar)            |
|--------------------------|-----------------------|
| Flow sensor (Kv 0.35)    | 0.04                  |
| Solenoid valve (Kv 0.25) | 0.08                  |
| 1 m of 1/4 in bore tube  | 0.012                 |
| Fittings (K = 6 in all)  | 0.012                 |
| Reactor ports            | 0.002                 |
| Reactor channel          | below 0.00001         |
| <b>Total</b>             | <b>0.15 (2.2 psi)</b> |

R8 is now worded to exclude the flow restrictor, which absorbs excess supply pressure by design. The unit drops 0.15 bar against 0.5 bar [G2]; with a Kv 0.5 sensor and a Kv 0.4 valve it would be 0.08 bar, and at the minimum supply of 2 bar, 1.85 bar remains for the restrictor and faucet [G3]. The margin covers the assumed flow coefficients, so **R8 is met**; the coefficients are still to be confirmed from the datasheets of the parts chosen.

The 57 x 10 mm window on its 51 mm seat carries a peak stress of 6.18 MPa at 8 bar, inside the 6.8 MPa design stress [G4]: **R7 is met**, with a 9 % margin (v0.1: 4.46 MPa on a 26 mm seat). Unprotected, a 16 bar water-hammer spike would raise it to 12.4 MPa. With the pressure limiter set at 4 bar the window carries 3.09 MPa static, and a transient doubling to 8 bar gives 6.18 MPa [G5]: **R17 is met** as an installation requirement. The 8 mm sensor window sees 1.69 MPa and the tube hoop stress is 8.3 MPa [G6]. Each cap carries an end load of 2,655 N; four M5 316 rods take 664 N each, 47 MPa in the thread [G7].

## ■ H. Size and I. Cost (R14, R16)

The unit is 277 x 100 x 322 mm without the adapter, inside the 350 x 150 x 350 mm envelope [H1]: **R14 is met**.

The 16-line BOM totals \$338.00 against the `budget_usd` of \$225, \$113.00 over [I1]: **R16 is not met**. The largest lines are the LEDs (\$54), the high-reflectance PTFE liner (\$45) and the machined 316 lower cap (\$42) [I2]. The rise from \$249.00 comes from the larger reactor: the liner, window, caps, tube and sink all grow with the 50 mm bore.

## ■ J. Flow limits and options

Table 5. Flow limits for this reactor and the options against R3.

| Case                                 | Result                                                                                                          |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Six LEDs, 90 %/cm                    | 40 mJ/cm <sup>2</sup> (laminar) up to 1.60 L/min; 48 mJ/cm <sup>2</sup> (Met with margin) up to 1.30 L/min [J1] |
| Same reactor at the former 2.0 L/min | 32.7 (laminar) to 45.9 (plug) mJ/cm <sup>2</sup> [J2]                                                           |
| Option B (R3 at 70 %/cm)             | 14 LEDs, 30 W of LED power [J3]; R9 would fail                                                                  |
| Option C (R3 at 70 %/cm)             | Six LEDs meet 40 mJ/cm <sup>2</sup> below 0.48 L/min [J4]; proportional valve, later upgrade under D1           |

The 1.2 L/min design flow sits inside the 1.30 L/min limit for the 20 % margin. Filling a 1 L bottle takes about 50 s.

## ■ K. Requirement status

Table 6. Requirement status at TRL 3 (LMF-REQ-001 v0.4). Not met items first.

| ID  | Target                                                  | Value (tag)                                                                                   | Status                          |
|-----|---------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------|
| R3  | RED 40 mJ/cm <sup>2</sup> or more at 1.2 L/min, 70 %/cm | 17.4 to 21.8 mJ/cm <sup>2</sup> [B6]                                                          | <b>Not met</b> (accepted by D1) |
| R16 | Parts cost \$225 or less; no custom PCB                 | \$338.00 [I1]; module-based electronics                                                       | <b>Not met</b>                  |
| R4  | Dose monitor; alarm and valve closed within 1 s         | Wall signal 16.3 nA at 90 %/cm, 0.38 nA at 70 %/cm; alarm below about 89 %/cm [C1, C4]        | At risk                         |
| R15 | Window and LED head replaced in 15 min                  | Four tie rods and two push-fit ports; needs a build to time                                   | Not verifiable at TRL 3         |
| R1  | 1.2 L/min, restrictor                                   | Restrictor; sensor range covers it [A3]                                                       | Met                             |
| R2  | RED 40 mJ/cm <sup>2</sup> or more at 1.2 L/min, 90 %/cm | 51.9 (laminar) to 76.3 (plug) mJ/cm <sup>2</sup> [B5]                                         | Met                             |
| R5  | LEDs on within 0.5 s above 0.3 L/min; 5 s run-on        | 0.24 s worst case [D2]                                                                        | Met                             |
| R6  | No mercury; full output in 0.1 s                        | LEDs, microsecond rise                                                                        | Met                             |
| R7  | Window 6.8 MPa or less at 8 bar                         | 6.18 MPa [G4]                                                                                 | Met                             |
| R8  | 0.5 bar or less at 1.2 L/min, excluding the restrictor  | 0.15 bar, on assumed Kv [G2]                                                                  | Met                             |
| R9  | 25 W flowing, 0.5 W standby                             | 22.6 W, 0.21 W [E1, E2]                                                                       | Met                             |
| R10 | LED board 50 °C or less; cut-back above 50 °C           | 37.1 to 44.8 °C steady [F3]                                                                   | Met                             |
| R11 | Food-contact wetted parts; no UV-C on plastics but PTFE | 316 lower cap; PTFE liner, top disc and 316 insert shield the acetal upper cap                | Met                             |
| R12 | No UV-C outside the unit; interlocks                    | Metal and PTFE light path; sensor window sealed by the photodiode saddle; interlock by design | Met                             |
| R13 | 24 V DC only at the unit                                | Certified adapter and input fuse                                                              | Met                             |
| R14 | 350 x 150 x 350 mm or less                              | 277 x 100 x 322 mm [H1]                                                                       | Met                             |
| R17 | Pressure limiter at 4 bar or less upstream              | Window 6.18 MPa at twice the setting [G5]                                                     | Met                             |

Counts: 13 met, 2 not met, 1 at risk, 1 not verifiable at TRL 3 [K1].

## ■ Numbers changed from v0.1

Table 7. v0.1 (25 mm bore, 2.0 L/min) against v0.2 (LMF-DDR-002 design).

| Quantity                                  | v0.1                                               | v0.2                                                     |
|-------------------------------------------|----------------------------------------------------|----------------------------------------------------------|
| Bore, channel, window                     | 25 mm, 246 mm, 32 x 6 mm                           | 50 mm, 242 mm, 57 x 10 mm [A1, G4]                       |
| Design flow                               | 2.0 L/min                                          | 1.2 L/min [A3]                                           |
| Liner reflectance (assumed)               | 0.80                                               | 0.95                                                     |
| UV-C absorbed by water at 90 %/cm         | 26.1 %                                             | 54.0 % [B3]                                              |
| RED at 90 %/cm                            | 14.7 to 19.0 mJ/cm <sup>2</sup> (R2 not met)       | 51.9 to 76.3 mJ/cm <sup>2</sup> (R2 met) [B5]            |
| RED at 70 %/cm                            | 6.3 to 7.4 mJ/cm <sup>2</sup>                      | 17.4 to 21.8 mJ/cm <sup>2</sup> [B6]                     |
| Dose sensor signal at 90 %/cm and 70 %/cm | 2.4 nA, unresolved                                 | 16.3 nA, 0.38 nA [C1]                                    |
| Worst-case switch-on                      | 0.46 s                                             | 0.24 s [D2]                                              |
| Pressure drop                             | 0.42 bar                                           | 0.15 bar [G2]                                            |
| LED board, nominal                        | 48.9 °C                                            | 40.3 °C [F3]                                             |
| Window stress at 8 bar                    | 4.46 MPa                                           | 6.18 MPa [G4]                                            |
| Energy                                    | 3.1 kWh/year                                       | 3.8 kWh/year [E3]                                        |
| Envelope                                  | 235 x 72 x 322 mm                                  | 277 x 100 x 322 mm [H1]                                  |
| Parts cost                                | \$249.00                                           | \$338.00 [I1]                                            |
| Requirements                              | 8 met, 4 at risk, 3 not met, 1 not verifiable (16) | 13 met, 1 at risk, 2 not met, 1 not verifiable (17) [K1] |